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Fine-Tuning of LLMs

System Prompts, Rule Setting, Knowledge Bases & Adaptation

Agenda

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Foundations

Understanding the difference between Prompting, RAG, and Fine-Tuning.

Prompt Engineering

Mastering System Prompts, Rules, and Few-Shot techniques.

Knowledge Base

Integrating external data through Retrieval Augmented Generation (RAG).

Fine-Tuning

When and how to update

Strategy

Decision frameworks for

Q&A

Addressing common

The Goal of Optimization

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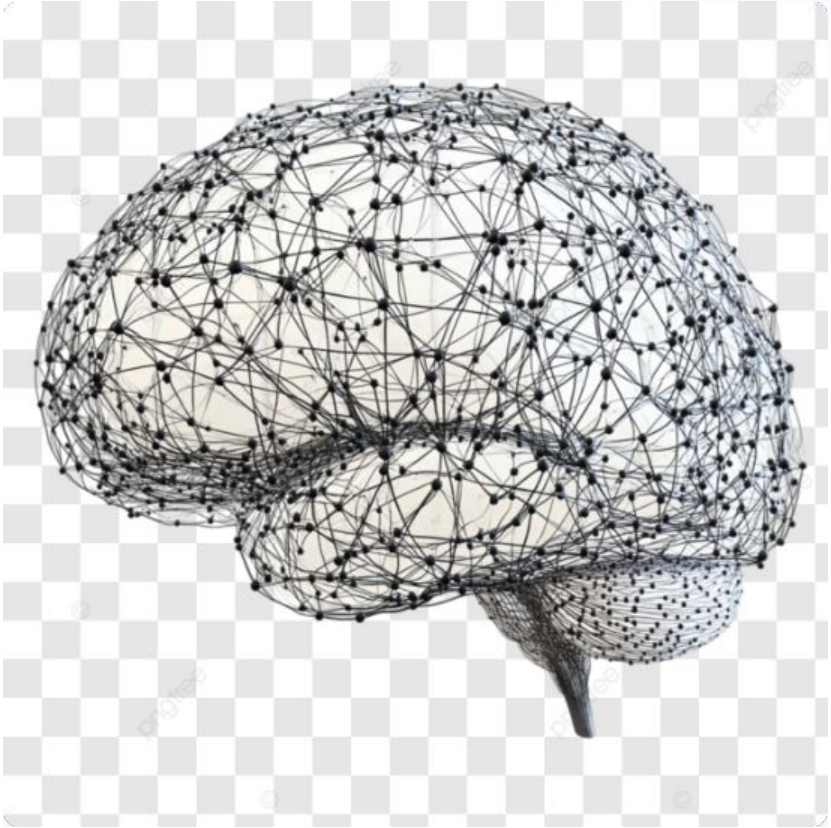
Out of the box, Large Language Models (LLMs) are generalists. To build effective applications, we must guide them.

- **Accuracy:** Reducing hallucinations and errors.
- **Consistency:** Ensuring strictly formatted outputs.
- **Context:** Providing domain-specific knowledge.
- **Tone:** Aligning with brand voice or persona.



The Optimization Spectrum

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1. Prompting

"In-Context Learning"

Giving instructions and examples in the input window. No model changes.

⚡Fastest to implement

2. RAG

"Retrieval Augmented Generation"

Injecting relevant knowledge from a database into the prompt dynamically.

📚Best for knowledge

3. Fine-Tuning

"Weight Adaptation"

Training the model on a dataset to permanently alter its behavior.

🏋Best for complex behaviors

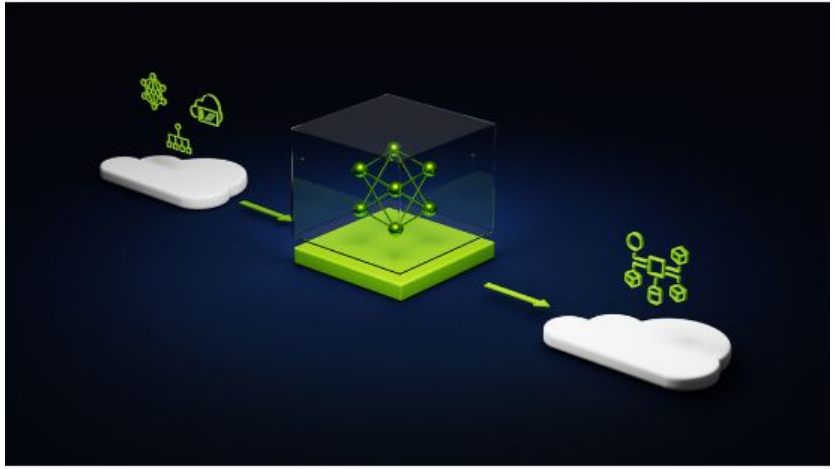
The Context Window

The "Working Memory"

Everything in Prompting and RAG happens within the **Context Window**.

This is the limit of text the model can consider at once. It includes:

- System Instructions
- Conversation History
- User Inputs
- Retrieved Knowledge (RAG)



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Part 2: Prompt Engineering

System Prompts, Rules, and Constraints

Anatomy of a Prompt

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Modern chat models use a specific structure of messages to guide behavior.

System

Sets the global behavior, persona, and strict rules. The "God Mode" instruction.

User

The specific task, question, or input data provided by the end-user.

Assistant

The model's response. Can be pre-filled to guide the format (few-shot).

System Prompt: Setting the Persona

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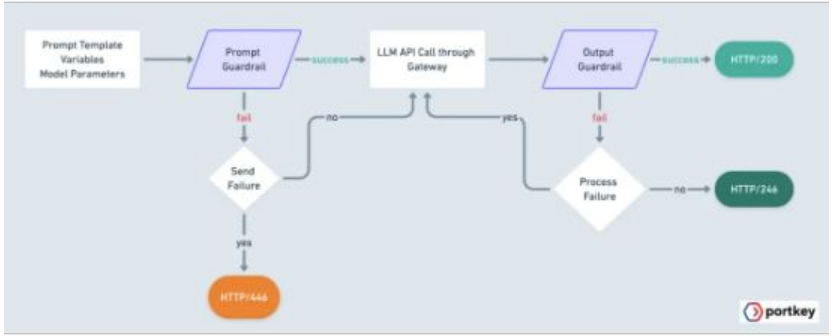
Defining "Who"

The System Prompt is the most powerful tool for "Soft Fine-Tuning". It stays persistent across the conversation.

Use it to define:

- **Role:** "You are a senior python engineer."
- **Tone:** "Be concise and professional."
- **Audience:** "Explain like I am five."



Setting Rules & Constraints

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Directives vs. Negative Constraints

Explicitly stating what **NOT** to do is often as important as what to do.

```
SYSTEM:
You are a customer support agent.

RULES:
1. NEVER invent information. If unsure, say "I don't know".
2. Always answer in less than 3 sentences.
3. Do not mention competitors by name.
4. Always end with "How else can I help?"
```

Enforcing Output Formats

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Structured Data

LLMs are excellent at transforming unstructured text into structured data (JSON, CSV, Markdown) if explicitly asked.

Technique: Provide a JSON schema in the system prompt.

```
SYSTEM: You are a data extractor. Output ONLY JSON.
Schema: { "customer_sentiment": "positive" |
"negative", "key_issues": ["list", "of", "issues"],
"urgency": 1-10 }
```

Zero-Shot vs. Few-Shot Prompting

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Zero-Shot

Asking the model to perform a task without examples.

"Classify this tweet: 'I loved the service!'"

Good for simple, common tasks.

Few-Shot (In-Context)

Providing 1-3 examples within the prompt to guide style and logic.

"Tweet: 'Worst day ever.' → Sentiment: Negative
Tweet: 'It was okay.' → Sentiment: Neutral
Tweet: 'I loved it!' → Sentiment: ?"

Drastically improves complex performance.

Chain of Thought (CoT)

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Encouraging Reasoning

For complex logic or math, asking the model to "think step-by-step" improves accuracy significantly.

Standard Prompt:

"Roger has 5 tennis balls. He buys 2 cans of tennis balls. Each can has 3 balls. How many does he have?"

Model: 11 (Sometimes fails)

CoT Prompt:

"...Let's think step by step."

Model: Roger starts with 5. 2 cans * 3 balls = 6 balls. 5 + 6 = 11.

Handling Hallucinations

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Hallucination is when the model confidently generates false information. Use these rules to mitigate it:

- **The "I Don't Know" Rule:** Explicitly instruct: "If the answer is not in the context, state 'I do not have that information'."
- **Source Citing:** "Please cite the specific section of the text used to answer."
- **Reduce Creativity:** Set `temperature` parameter to 0 for factual tasks.

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Part 3: Providing Knowledge

Retrieval Augmented Generation (RAG)

The Knowledge Gap

Why models fail at facts

LLMs are frozen in time. They only know what they learned during pre-training.

- They don't know your private company data.
- They don't know news from yesterday.
- They can hallucinate details about specific documents.

What is RAG?

Retrieval Augmented Generation

Instead of retraining the model, we give it an "Open Book" test.

1. Retrieve

Search your database for relevant documents based on the user's question.

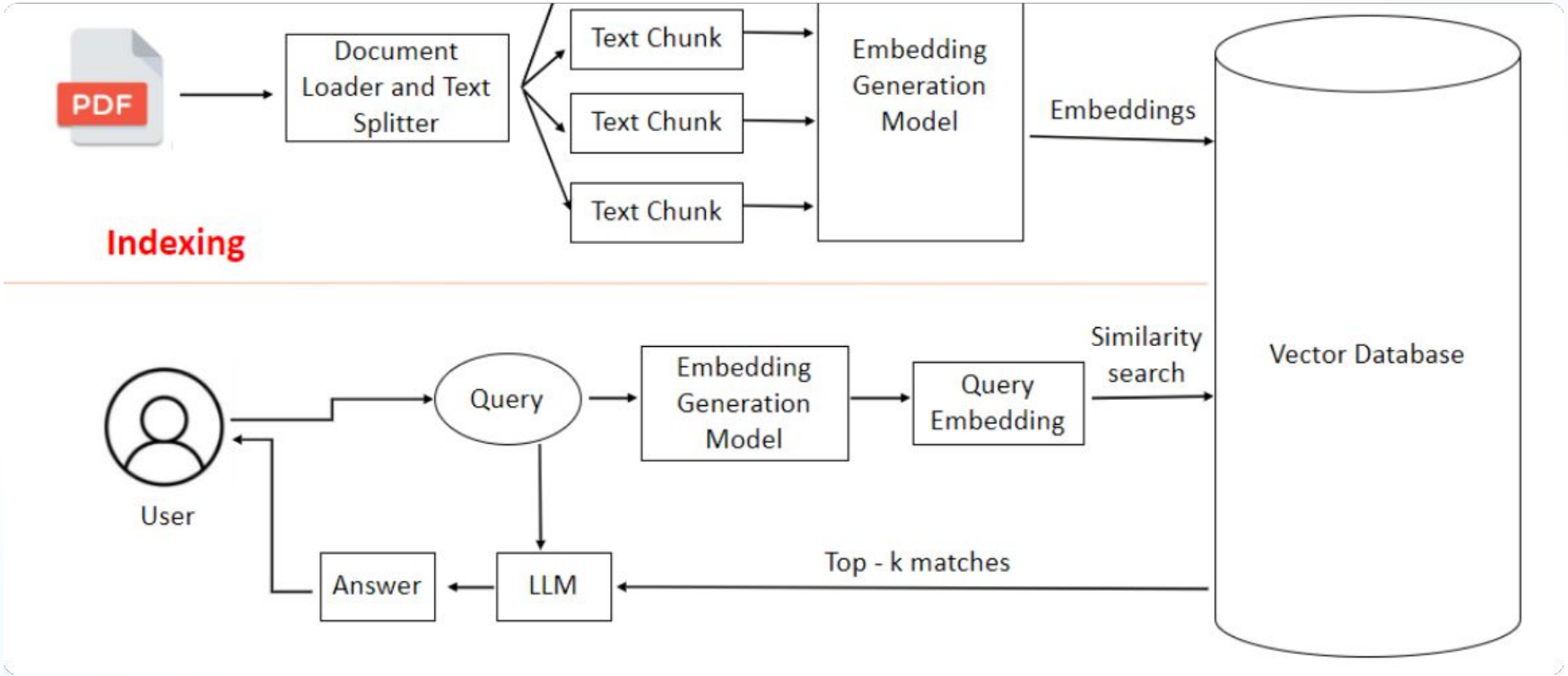
2. Augment

Paste those documents into the System Prompt as "Context".

3. Generate

The LLM answers the question using ONLY the provided context.

RAG Architecture



RAG vs. Fine-Tuning for Knowledge

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Feature	Fine-Tuning	RAG (Knowledge Base)
Knowledge Update	Slow (Retraining required)	Instant (Update database)
Accuracy	Prone to Hallucination	High (Grounded in facts)
Citations	Difficult	Easy (Direct references)
Privacy	Data baked into model	Data separated & controlled

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Part 4: Fine-Tuning

Adapting the Model Weights

What is Fine-Tuning?

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Beyond Prompting

Fine-tuning involves training the model on a dataset of examples to permanently adjust its internal parameters.

It's not usually for adding *knowledge*, but for teaching *behavior*, *format*, or *style*.

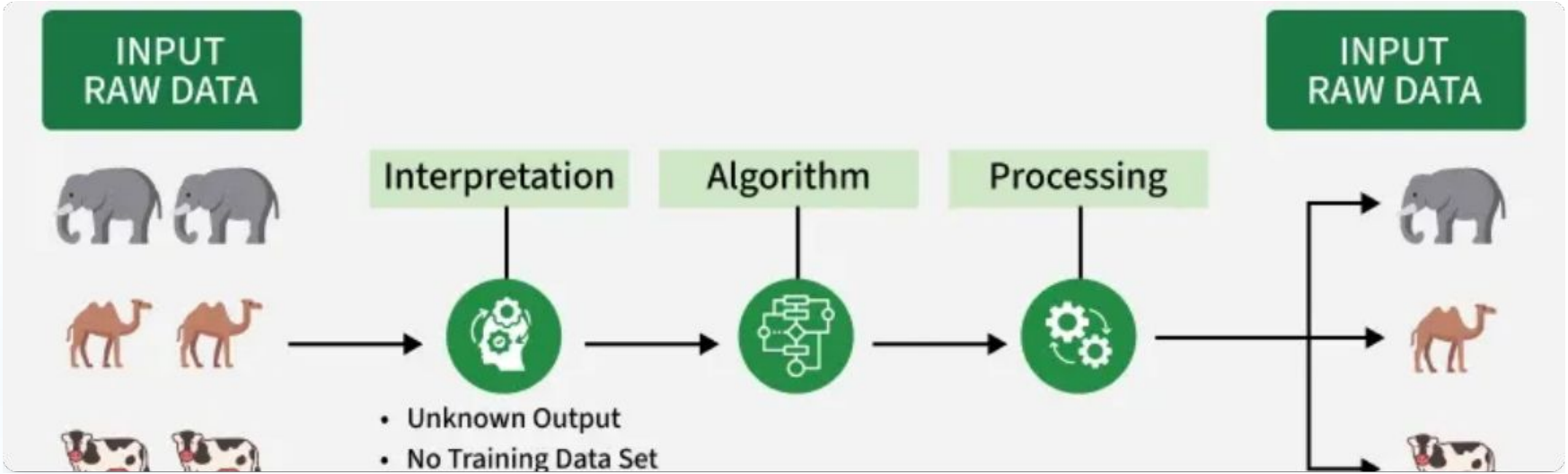
Example Use Cases

- Teaching a model to speak specialized legalese.
- Optimizing a small model (e.g., Llama-7B) to perform as well as GPT-4 on a specific task.
- Reducing token usage by removing the need for long instructions.

Instruction Tuning

The most common form of fine-tuning today is **Instruction Tuning**.

We train the model on pairs of (Instruction, Output) so it learns to follow instructions better.



The Dataset (JSONL)

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To fine-tune, you need a high-quality dataset. Quality matters more than quantity.

```
{"messages": [ {"role": "system", "content": "You are a sarcastic chatbot."}, {"role": "user", "content": "What is the capital of France?"}, {"role": "assistant", "content": "It's Paris, obviously. Did you skip geography?"} ]} {"messages": [ {"role": "system", "content": "You are a sarcastic chatbot."}, {"role": "user", "content": "Help me with math."}, {"role": "assistant", "content": "Fine. But I'm not a calculator."} ]}
```

PEFT & LoRA

Parameter-Efficient Fine-Tuning

Full fine-tuning updates all 7B+ parameters, which is expensive.

LoRA (Low-Rank Adaptation) freezes the main model and trains only a tiny adapter layer (1% of parameters).

- Runs on consumer GPUs.
- Fast training (hours vs days).
- Modular (swap adapters for different tasks).

Prompting vs. Fine-Tuning Matrix

Scenario	Recommended Approach
Need new knowledge (News, Company Data)	**RAG (Retrieval)**
Need specific output format (JSON, Code)	**Prompting (Few-Shot)**
Need specific tone/style consistently	**Fine-Tuning**
Prompt is too long / expensive	**Fine-Tuning**
Need to fix occasional reasoning errors	**Chain of Thought Prompting**

Summary Strategy

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1

Start with Prompting

Use System Prompts and Few-Shot examples. This solves 80% of problems.

2

Add RAG

If you need external data or facts, connect a Knowledge Base.

3

Fine-Tune Last

Only fine-tune if you need to reduce latency, cost, or deeply bake in a behavior.

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Questions?

Ask about Prompting, RAG, or Fine-Tuning

Common Questions

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Does Fine-Tuning stop hallucinations?

Not necessarily. In fact, it can sometimes increase them if the training data contains errors. RAG is generally better for factuality because it grounds the answer in retrieved text.

How many examples do I need for Fine-Tuning?

For style transfer, as few as 50-100 high-quality examples can work. for complex reasoning, you might need 1,000+. Quality > Quantity.

Common Questions

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Can I combine RAG and Fine-Tuning?

Yes! This is the state-of-the-art. You fine-tune a model to be really good at using retrieved context (Instruction Tuning) and then use RAG to feed it the latest data.

What is the "System Prompt" length limit?

It shares the same context window as the user input. If your system prompt is huge (e.g., 20 pages of rules), you have less space for the conversation.

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Thank You

Optimizing LLMs is an iterative process.

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